Notes: Dicks & Fulghieri (RFS, 2025)
Uncertainty, Contracting, and Beliefs in Organizations

Contents

1 Big picture

- **Question.** How does uncertainty, or ambiguity, change optimal incentive contracts inside a multidivisional firm?
- **Standard benchmark.** In the Holmstrom-Milgrom style principal-agent model, incentive pay should load on informative performance measures and should hedge risk. If two divisions have positively correlated cash flows, the contract should use relative performance, because subtracting the other division reduces common noise.
- **New ingredient.** Division managers and headquarters are uncertainty averse. They do not know the true productivity distribution and evaluate contracts using worst-case beliefs from a core beliefs set.
- **Core mechanism.** Incentive contracts affect beliefs. Pay based only on a manager's own division makes the manager pessimistic about own division productivity. Cross-pay, such as equity-based pay or relative performance, hedges uncertainty and can make the manager less pessimistic.
- **Why beliefs matter for moral hazard.** Effort incentives depend on perceived marginal productivity. If the manager believes own productivity is low, a given pay-performance sensitivity produces less effort. Headquarters must then raise incentive intensity, which is costly.
- **Disagreement channel.** Uncertainty aversion creates endogenous disagreement between HQ and managers. A contract valued by HQ may be valued at a discount by the manager because each side has different exposure to uncertainty.
- **Main result.** Cross-division exposure can be optimal even when division cash flows are uncorrelated. This violates the standard informativeness principle because the cross-pay does not add information about the manager's action; it improves beliefs by hedging uncertainty.
- **Equity-pay result.** When uncertainty is high enough, equity-based pay can dominate relative-performance pay. This helps explain why equity-based incentive contracts are common and why explicit relative-performance contracts are rare.
- **HQ uncertainty matters.** If HQ is itself uncertainty averse, the optimal contract trades off uncertainty hedging for managers against disagreement and uncertainty discounts faced by HQ.
- **Internal hedges.** The model favors internal hedges, such as other divisions' cash flows, over external benchmarks, because internal hedges affect both the manager's beliefs and HQ's valuation.
- **Empirical implications.** Firms facing high uncertainty, young firms, and firms where managerial beliefs are hard to align should use more aggregate or equity-based pay and less relative-performance pay.
- **Takeaway.** Optimal incentive contracts are not only about risk sharing and action informativeness. They are also belief-management devices.

2 Model environment and economic primitives

2.1 Organization and timing

- The firm has headquarters and two divisions, A and B.
- Division manager $d \in \{A, B\}$ chooses unobservable effort a_d.
- Division cash flow is Y_d, and effort affects the mean of cash flow through division productivity q_d.
- HQ commits to compensation contracts at date 0.
- Managers accept contracts and simultaneously choose effort.
- Cash flows are realized and compensation is paid at date 1.

Interpretation: The model is a moral-hazard problem inside an organization. The friction is not that HQ cannot observe output; it can. The friction is that HQ cannot observe effort and cannot perfectly pin down productivity beliefs under uncertainty.

2.2 Cash flows

Division cash flows have joint normal distribution

$$Y = (Y_A, Y_B) \sim N(\mu, \Sigma), \tag{1}$$

with

$$\mu_d = a_d q_d, \qquad d \in \{A, B\}. \tag{2}$$

Cash flows are homoscedastic with variance σ^2 and correlation coefficient ρ.

Interpretation: Effort changes expected cash flow but not the variance-covariance matrix. This isolates incentive and belief effects from technology effects.

2.3 Effort costs and utility

Effort cost is quadratic:

$$c_d(a_d) = \frac{1}{2\theta_d} a_d^2, \tag{3}$$

where θ_d measures the efficiency of effort. Division managers have CARA utility:

$$U(w_d) = -e^{-rw_d}, \tag{4}$$

where r is the coefficient of absolute risk aversion.

Interpretation: The CARA-normal structure keeps the model analytically tractable. It allows the paper to separate risk premia, incentive intensity, and belief distortions.

2.4 Linear incentive contracts

The contract for division manager d is linear in both division cash flows:

$$w_d(Y) = s_d + \beta_d Y_d + \gamma_d Y_{d'}, \qquad d' \neq d. \tag{5}$$

Here:

- s_d is fixed pay;
- β_d is own-division pay-performance sensitivity;
- γ_d is cross-division exposure.

Interpretation: If $\gamma_d > 0$, the contract has an aggregate or equity-like component. If $\gamma_d < 0$, the contract has a relative-performance component. If $\gamma_d = 0$, it is pure pay-for-performance on own division.

2.5 Uncertainty aversion and multiple priors

Agents evaluate contracts using minimum expected utility:

$$U = \min_{p \in \mathcal{P}} \mathbb{E}_p[U(w)]. \tag{6}$$

The core beliefs set is based on relative entropy. For distributions $\widehat{P}$ and P with densities $\hat{p}$ and p,

$$R(\widehat{P} \mid P) = \int \hat{p}(x) \ln\left(\frac{\hat{p}(x)}{p(x)}\right) \mathrm{d}x. \tag{7}$$

The admissible set is

$$\mathcal{P}(P) = \{\widehat{P} : R(\widehat{P} \mid P) \leq \eta_P\}. \tag{8}$$

Interpretation: The parameter η_P controls the size of the uncertainty set. Higher η_P means the agent considers more distorted beliefs plausible and is less confident in the reference model.

2.6 Uncertainty hedging

The key mathematical property is that randomizing or combining uncertain exposures can be strictly preferred under multiple priors:

$$\alpha \min_{p \in \mathcal{P}} \mathbb{E}_p[U(y_1)] + (1-\alpha) \min_{p \in \mathcal{P}} \mathbb{E}_p[U(y_2)] \leq \min_{p \in \mathcal{P}} \mathbb{E}_p[\alpha U(y_1) + (1-\alpha) U(y_2)]. \tag{9}$$

Interpretation: This is the ambiguity analogue of diversification. By exposing the manager to multiple divisions, HQ can reduce worst-case pessimism about any single division.

3 Main models and theoretical results

3.1 No-uncertainty benchmark

In the standard benchmark, HQ is risk neutral and managers are risk averse but not uncertainty averse. The optimal contract has

$$\gamma_d^* = -\rho\beta_d^*. \tag{10}$$

Interpretation: This is standard risk hedging. If division cash flows are positively correlated, the contract subtracts the other division's performance. If they are negatively correlated, the contract adds it. Cross-pay is used only because it hedges risk.

3.2 Belief distortion from own-performance pay

When a manager is paid mainly on own division output, the manager chooses worst-case beliefs about own productivity. Let $\widehat{q}_d^i$ denote agent i's belief about productivity of division d. A manager paid on own output tends to choose

$$\widehat{q}_d^d < q_d. \tag{11}$$

Interpretation: Pessimistic beliefs reduce the perceived marginal benefit of effort. HQ must pay a stronger incentive slope to induce the same effort.

3.3 Cross-pay and the uncertainty-hedging ratio

The paper defines the relative exposure of manager d to the two divisions through a hedging ratio, written schematically as

$$H_d \equiv \frac{\gamma_d a_{d'} q_{d'}}{\beta_d a_d q_d}. \tag{12}$$

The exact belief case depends on whether this relative exposure is small, intermediate, or large.

Interpretation: The hedging ratio summarizes whether the manager's payoff is dominated by own-division uncertainty, balanced across divisions, or dominated by other-division exposure.

3.4 Effort provision

Given contract (β_d, γ_d) and manager belief $\widehat{q}_d^d$, equilibrium effort satisfies

$$a_d = \beta_d \theta_d \widehat{q}_d^d. \tag{13}$$

Interpretation: This is the paper's key moral-hazard-belief link. Incentive intensity matters, but the belief-adjusted productivity term also matters. Cross-pay can increase effort by improving $\widehat{q}_d^d$ even without changing β_d.

3.5 Uncertainty-neutral HQ

If HQ is not uncertainty averse but division managers are, cross-division exposure can be optimal even when cash flows are uncorrelated. A key result is that HQ uses cross-pay to make managers less pessimistic and thereby reduce incentive costs.

In the risk-neutral manager case, the paper obtains contracts with

$$|\gamma_d| = \xi_d \beta_d, \tag{14}$$

where ξ_d captures the uncertainty-hedging role of cross-pay.

Interpretation: The sign of γ_d is not central when there is no risk-sharing motive. What matters is that the manager's compensation depends on more than one uncertain source.

3.6 Risk and uncertainty hedging

When managers are both risk averse and uncertainty averse, the optimal cross-pay must trade off two motives:

- **Risk hedging:** use negative exposure to positively correlated cash flows.
- **Uncertainty hedging:** use cross exposure to improve worst-case beliefs.

The first-order condition equates the incentive value and risk cost of own-division and cross-division exposure:

$$\beta_d a_d q_d + r\sigma^2 \beta_d^2 = |\gamma_d| a_{d'} q_{d'} + r\sigma^2 \gamma_d^2. \tag{15}$$

Interpretation: This condition shows why the standard informativeness principle no longer fully governs contract design. Cross-pay has value through belief management, not only through information or risk hedging.

3.7 Corollary: uncertainty and cross-pay

For symmetric divisions and uncorrelated cash flows, there is a threshold $\bar{\eta}$ such that:

- if $\eta \leq \bar{\eta}$, the optimal contract resembles the standard risk-hedging contract;
- if $\eta > \bar{\eta}$, the optimal contract uses full cross exposure, with $|\gamma| = \beta$.

Interpretation: Low uncertainty leaves the standard model approximately intact. High uncertainty changes the objective: HQ uses cross-pay to manage beliefs.

3.8 Uncertainty-averse HQ

When HQ is also uncertainty averse, HQ's own beliefs become endogenous. The optimal contract must now consider:

- the manager's pessimism about own productivity;
- HQ's pessimism about division productivity;
- the disagreement discount between manager and HQ valuations;
- the effect of cross-pay on effort and participation constraints.

Interpretation: HQ uncertainty can make relative performance less attractive. A contract that hedges manager risk may create disagreement or make HQ pessimistic about the value of compensation payments.

3.9 Flexible leadership and equity contracts

With flexible leadership, high uncertainty can lead HQ to choose equity-based pay:

$$\gamma = \beta. \tag{16}$$

This can occur even when cash flows are positively correlated, where the standard benchmark would recommend

$$\gamma = -\rho\beta. \tag{17}$$

Interpretation: High uncertainty can reverse the standard relative-performance prediction. The firm sacrifices risk hedging to coordinate beliefs and reduce the uncertainty discount.

3.10 Internal versus external hedges

The paper extends the model by allowing contracts to depend on an external hedging variable C. It shows that HQ prefers internal hedges to external hedges in relevant cases.

Interpretation: Internal hedges are informationally and organizationally special because they affect both sides of the internal contracting problem. They can align beliefs between managers and HQ more effectively than a purely external benchmark.

4 Conceptual identification and empirical content

4.1 What the model explains

The model helps explain several empirical regularities:

- lower-level managers and employees often receive aggregate or equity-based compensation;
- explicit relative-performance contracts are relatively rare;
- firms facing more uncertainty may rely more on equity-based pay;
- incentive plans sometimes include "pay-for-luck," even though luck is not informative about effort;
- business groups and multidivisional firms may use broad internal performance measures to coordinate beliefs.

Interpretation: These facts look puzzling in the standard risk-sharing model. They become more natural if aggregate pay hedges uncertainty and manages beliefs.

4.2 Why the result is not merely risk sharing

- The most striking result occurs even when cash flows are uncorrelated.
- In that case, cross-pay does not hedge risk in the usual covariance sense.
- Its role is to reduce worst-case pessimism and disagreement.

Interpretation: The model adds a separate contracting motive. A performance measure can be useful even if it is not informative about effort and does not reduce variance.

4.3 Where to look empirically

The paper suggests that one should examine:

- young firms versus mature firms;
- high-uncertainty industries versus low-uncertainty industries;
- firms with dispersed or hard-to-measure divisional productivity;
- equity-based pay for divisional managers and rank-and-file employees;
- the absence or presence of relative performance evaluation;
- benchmarking in firms with different degrees of uncertainty.

Interpretation: The empirical content is comparative statics. As uncertainty rises, the model predicts more aggregate pay and less pure relative-performance pay.

4.4 Limitations of the theoretical design

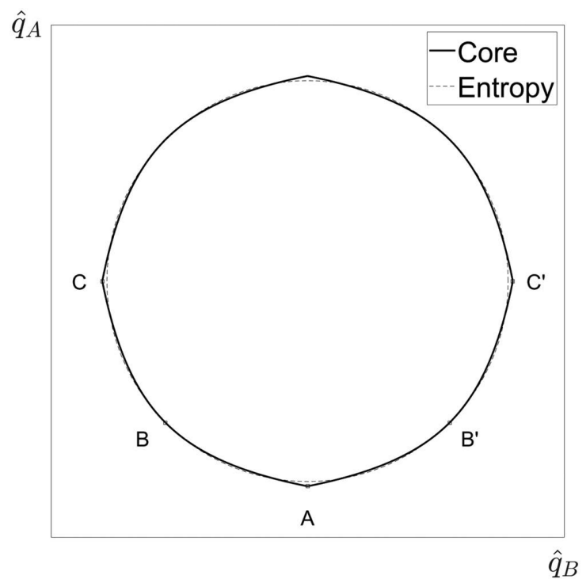
- The paper is primarily theoretical rather than empirical.
- It uses a two-division structure for tractability.
- Contracts are linear.
- Uncertainty is modeled through relative entropy and multiple priors.
- Managers and HQ share the same reference beliefs; disagreement is endogenous rather than assumed.
- Limited liability and richer organizational features are not the central mechanism.

Interpretation: The model is intentionally stylized. Its strength is that it isolates a clean mechanism: contracts change beliefs by changing uncertainty exposure.

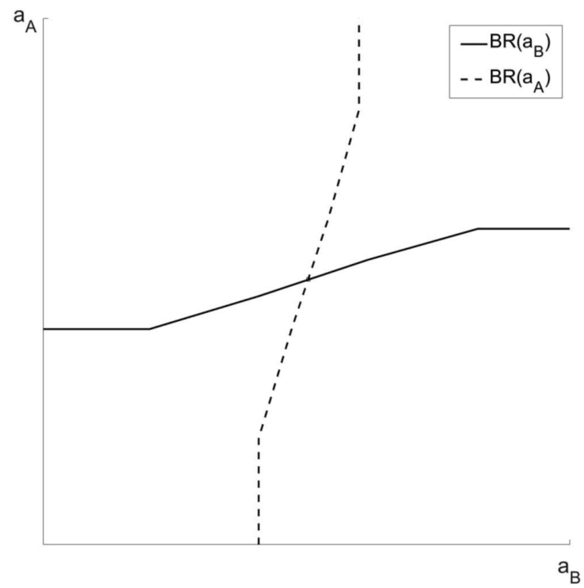
5 Figures and exhibit interpretation

This paper has no empirical tables. The visual content consists of theoretical figures. The figures below are directly cropped from the original paper and grouped compactly to speed up reading.

5.1 Figure 1 and Figure 2: core beliefs and effort interaction



(a) Figure 1: core beliefs set.



(b) Figure 2: best responses and equilibrium effort.

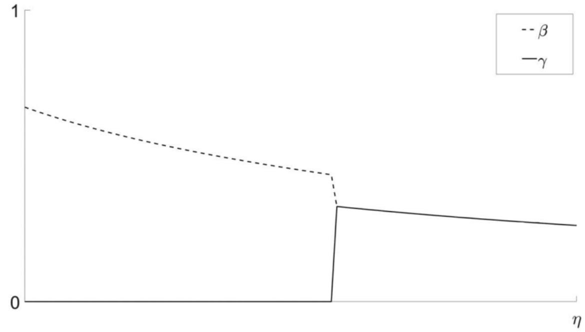
Read:

- Figure 1 shows the core beliefs set under relative entropy. The boundary is smooth and convex.
- Points in the figure correspond to different cases of how a manager distorts beliefs depending on own-pay and cross-pay exposure.
- When own-pay dominates, the manager is pessimistic about own productivity.
- When cross-pay balances own-pay, the manager can hold less pessimistic beliefs about both divisions.
- Figure 2 shows best response functions. Cross-pay changes beliefs, which shifts effort responses.

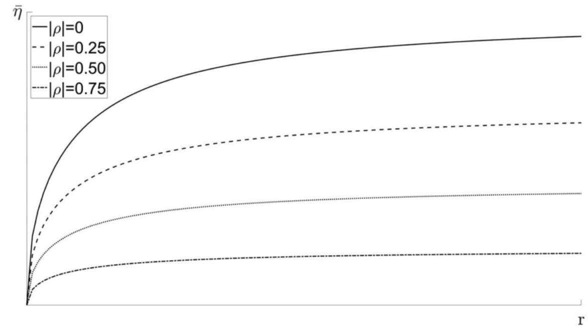
Economic message: These figures show the two building blocks of the paper. Figure 1 shows how uncertainty exposure determines worst-case beliefs. Figure 2 shows why those beliefs matter: they enter the effort best response and can create a positive feedback across divisions.

Interpretation: In the standard model, another division's payoff exposure affects a manager only through risk. Here it affects the manager's perceived productivity. That makes cross-pay a belief-management tool. Figure 2 is especially important because it demonstrates that the model is not just about compensation valuation. Belief distortions feed directly into effort and therefore into real output.

5.2 Figure 3 and Figure 4: visionary leadership and the cross-pay threshold



(a) Figure 3: optimal contract with visionary leadership.



(b) Figure 4: cross-pay uncertainty threshold.

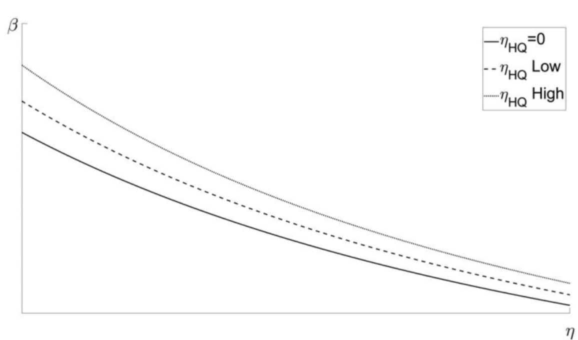
Read:

- Figure 3 plots own-performance sensitivity β and cross-division exposure γ as uncertainty rises.
- At low uncertainty, the contract keeps γ near the standard benchmark.
- Once uncertainty exceeds a threshold, the contract gives full exposure to the other division, so $\gamma = \beta$ in the uncorrelated case.
- Figure 4 plots the threshold $\bar{\eta}$ as a function of risk aversion and cash-flow correlation.
- Higher managerial risk aversion raises the cost of cross-pay, so the threshold for using it is higher.

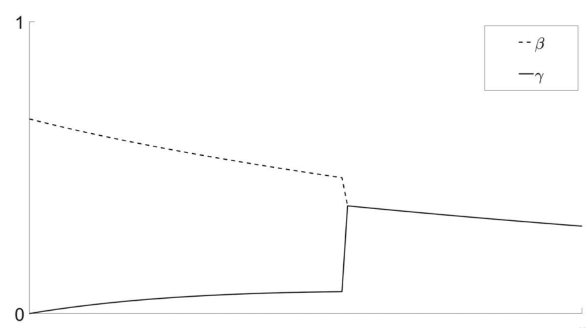
Economic message: These figures show when uncertainty hedging becomes strong enough to overturn standard incentive logic. Cross-pay is not always optimal; it becomes optimal when uncertainty is large enough relative to risk-sharing costs.

Interpretation: The threshold result is useful because it prevents the theory from predicting equity pay everywhere. In low-uncertainty settings, the classic informativeness principle still has force. In high-uncertainty settings, beliefs become the binding problem and aggregate/equity pay becomes attractive. Figure 4 also shows that risk aversion and ambiguity aversion are not the same force: risk aversion can make cross-pay more expensive even when uncertainty aversion makes it more useful.

5.3 Figure 5 and Figure 6: HQ uncertainty and flexible leadership



(a) Figure 5: pay-performance sensitivity under HQ uncertainty.



(b) Figure 6: flexible leadership with uncorrelated cash flows.

Read:

- Figure 5 shows how β changes when HQ is also uncertainty averse.
- If HQ uncertainty rises, pay-performance sensitivity can fall because HQ discounts the value of inducing effort through contracts exposed to uncertain cash flows.
- Figure 6 shows that with uncorrelated cash flows, rising uncertainty can make γ increase from zero to β.
- For sufficiently high uncertainty, pure equity-like contracts emerge even though the other division's cash flow is not informative about effort.

Economic message: HQ uncertainty adds a second layer. Contracts must not only motivate managers but also be valued by a potentially pessimistic principal.

Interpretation: Figure 6 is one of the clearest departures from the standard model. If cash flows are uncorrelated, risk-sharing theory would not put other-division output into the contract. The figure shows that ambiguity changes this prescription. The reason is not covariance; it is the shape of the belief set. Spreading the manager's payoff over two uncertain divisions reduces worst-case pessimism and improves effort incentives.

5.4 Figure 7: flexible leadership with positive correlation

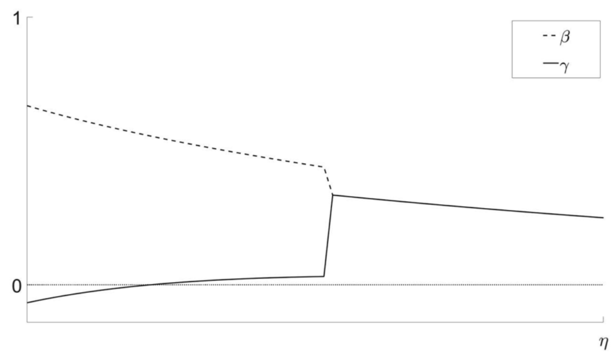


Figure 7: optimal contract with flexible leadership and positively correlated cash flows.

Read:

- With no uncertainty and positive correlation, the standard contract uses relative performance: $\gamma = -\rho\beta$.
- As uncertainty rises, γ increases toward zero and then becomes positive.
- At high uncertainty, the contract becomes equity-like: $\gamma = \beta$.
- The figure therefore shows a full reversal from relative-performance pay to equity-based pay.

Economic message: This figure explains why relative-performance pay may be rare in practice. Even when correlation says relative performance should hedge risk, uncertainty says aggregate pay may better align beliefs and reduce contracting costs.

Interpretation: Figure 7 is the paper's strongest practical prediction. Most firms face common shocks, so standard agency theory would predict more relative performance evaluation. The model shows that high uncertainty can overturn this prediction. This helps reconcile the observed prevalence of broad bonuses and equity pay with the theoretical puzzle that such pay exposes employees to firm-wide or market-wide shocks.

6 Short summary

- The paper studies incentive contracts in a multidivisional firm under uncertainty aversion.
- Division managers choose hidden effort, and HQ designs linear contracts based on own and other division cash flows.
- Agents use multiple priors and evaluate contracts under worst-case beliefs.
- Contract exposure determines beliefs; beliefs determine effort.
- Own-performance pay can make managers pessimistic about their own division productivity.
- Cross-pay can hedge uncertainty and make managers more optimistic, raising effort for a given incentive slope.
- This can make cross-division exposure optimal even when other-division performance is not informative about effort.
- When risk aversion is present, cross-pay trades off uncertainty hedging against risk exposure.
- When HQ is also uncertainty averse, disagreement and uncertainty discounts become central.
- High uncertainty can make equity-based pay optimal even when cash flows are positively correlated.
- Internal hedges can dominate external benchmarks because they help align internal beliefs.
- The model explains equity pay, aggregate bonuses, pay-for-luck, and limited use of relative-performance contracts.

7 Conclusion

7.1 Core conclusion

Dicks and Fulghieri argue that uncertainty changes the design of optimal incentive contracts because contracts shape beliefs. In a standard principal-agent model, performance measures are useful when they are informative about effort or when they hedge risk. In this paper, a performance measure can be useful even when it is not informative and even when it increases risk, because it can hedge uncertainty and reduce pessimistic belief distortions.

7.2 Economic intuition

The central intuition is that an uncertainty-averse manager evaluates a contract under a worst-case productivity belief. If compensation depends only on own-division output, the worst case is that own productivity is low. This weakens effort incentives. If compensation also depends on another division, the manager's payoff is spread over multiple uncertain sources. Under relative entropy, it may be too pessimistic to believe that all relevant productivity states are simultaneously bad. The contract therefore improves beliefs, and better beliefs make effort cheaper to induce.

7.3 Implications for incentive design

The paper suggests that broad incentive pay is not necessarily a mistake. Equity pay, firm-wide bonuses, and aggregate performance measures may appear inefficient if viewed only through risk sharing. But they may be efficient if they reduce ambiguity about the value of effort. This is especially relevant for lower-level managers and employees whose actions do not directly determine firm-wide performance. The model gives a reason why firm-wide pay can be useful even when the employee has little direct control over the whole firm.

7.4 Implications for relative-performance evaluation

The model also helps explain why relative-performance evaluation is rarer than standard theory predicts. If divisions or firms face common shocks, relative performance is a natural risk hedge. But relative-performance contracts may worsen disagreement or fail to provide uncertainty hedging. When ambiguity is high, the firm may prefer equity-like pay even in settings where correlation-based risk sharing recommends negative cross-exposure.

7.5 Implications for organizational design

The model connects compensation design to organizational structure. A multidivisional firm can use internal divisions as hedging instruments. Internal performance measures are not just accounting objects; they shape the belief environment of managers. This suggests that conglomerates, business groups, and firms with broad internal labor markets may be able to use internal pay links to coordinate beliefs and incentives.

7.6 Implications for leadership and corporate culture

The paper's language of belief alignment suggests a role for leadership. "Visionary" leadership can be interpreted as HQ having relatively confident beliefs and using contracts to bring managers' beliefs closer to its own. Under this interpretation, compensation and corporate culture are complements. Contracts provide monetary exposure; leadership provides a common narrative or reference model that reduces disagreement.

7.7 Implications for empirical work

The theory points to several empirical tests. Firms with higher uncertainty should rely more on equity-based or aggregate pay. Relative-performance contracts should be less common when uncertainty is high, even if cash-flow correlation is positive. Benchmarking should be more common when it helps reduce ambiguity rather than merely hedge risk. Tests should distinguish risk exposure from uncertainty exposure because they can have opposite implications for contract design.

7.8 Limitations

The paper is theoretical and stylized. It abstracts from limited liability, dynamic career concerns, detailed accounting rules, endogenous information acquisition, and many legal features of compensation contracts. It also uses the relative-entropy representation of ambiguity, which is powerful but specific. These limitations do not weaken the central mechanism, but they matter for mapping the model to real compensation data.

7.9 Research agenda

Future research can measure firm-level or division-level uncertainty and relate it to compensation breadth. Another direction is to examine whether broad incentive pay is more common when managers have heterogeneous forecasts or when productivity is hard to estimate. A third direction is to test whether mergers, internal capital markets, or business groups increase the use of internal cross-pay because they create better internal hedges.

7.10 Final takeaway

The main lesson is that contracts do more than allocate risk and incentives. They also allocate uncertainty. Once beliefs are endogenous, aggregate pay, equity pay, and pay-for-luck are not necessarily failures of governance. They may be deliberate tools for reducing disagreement and making managers confident enough to exert effort.